Section: 03

Group No: 07

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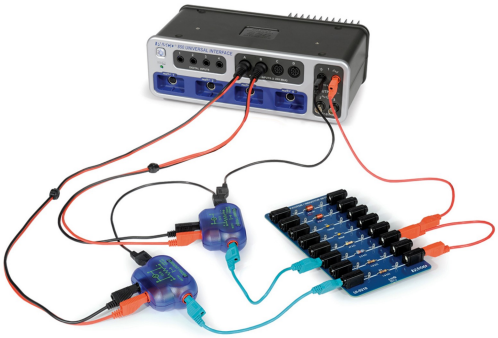
Kirchhoff's Circuit Laws

Equipment

Included:

1	Resistive/Capacitive/Inductive Network	UI-5210
3	Voltage Sensors	UI-5100
2	Current Probe	PS-2184
7	Short Patch Cords	SE-7123
Required but not included:		
1	850 Universal Interface	UI-5000



Introduction

Kirchhoff's Laws form the basis of all circuit analysis. Here we verify the laws for a resistive circuit using a DC input and for a time varying RC circuit.

Written by Chuck Hunt

Theory

Kirchhoff's Rules (sometimes called laws) state:

1. Junction Rule: the total current flowing into any point is zero at all times where we use the convention that current into a point is positive and current out of the point is negative.

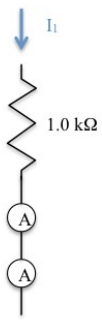
$$\sum I = 0$$

2. Loop Rule: the sum of the voltage drops around any closed loop must equal zero where the drop is negative if the voltage decreases and positive if the voltage increases in the direction that one goes around the loop.

$$\sum V = 0$$

Calibration Setup

Construct the circuit shown in the Circuit Diagram 1 or in Figure 1. A l000 W (±5%) resistor is connected in series with two Current Probes (the A with a circle around it [for ammeter] on the circuit diagram). A Voltage sensor is attached to each Current Probe as shown and then to the A and Analog inputs on the 850 Universal Interface. It is important to observe polarity by connecting red to red and black to black where possible. It is also important to keep track of which Current Probe is attached to which Analog input. You might attach a piece of tape to one of the Current Probes and mark it A.



Circuit Diagram 1

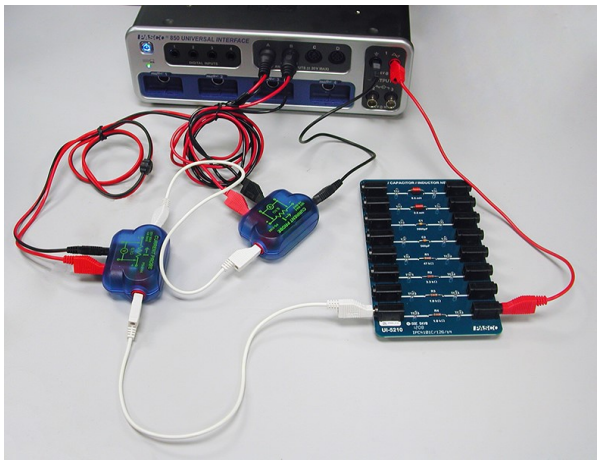


Figure 1: Ammeter Calibration

Calibration Procedure

The Current Probes (and the Output 1 current measurement) work by measuring the voltage drop across a small resistor (0.1 Wfor the Current Probes). Since the sensitivity is about 0.1 mA, this means the 850 Universal Interface must measure voltages of 0.01 mV. Noise can result in significant zero error. By averaging over several seconds we can achieve a precision of 0.1-0.2 mA, but with systematic zero errors that can be several milliamps. We can correct for this with a brief calibration procedure.

1. Click open the Signal Generator at the left of the screen. Set 850 Output 1 for a DC Waveform and a DC Voltage of 0 V. Click the On button.

2. Click Record (bottom left of screen).

3. Wait several seconds until the measured currents stop varying as the average becomes well defined. Click Stop.

4. Enter the three values in the first column of the table below.

5. Click Delete Last Run at the bottom of the screen.

6. In the Signal Generator panel, increase the voltage to 5 V and repeat. Then repeat for 10 V and 15 V. Turn Signal Generator off.

7. From the values in the table, calculate the average current correction for each ammeter and enter it in the "Current Correct" column of the table. Note that since we are using a 1000 Wresistor, the current should be 5.0 mA for a 5 V input and so on. Also notice that if all the values are high, the correction should be negative.

8. Open up the Calculator and put the values of the last column(current correct) according to the following examples: Ammeter 1(Current Correct) => Z; Ammeter A(Current Correct) => A; Ammeter B(Current Correct)=> B.

Output 1 Current

i1 ave (mA) <No Data Selected>

-.-mA

Current Probe A

iA ave (mA) <No Data Selected>

-.-mA

Current Probe B

iB ave (mA) <No Data Selected>

-.-mA

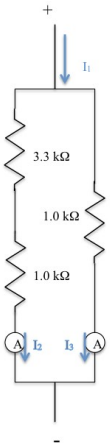
Table I: Ammeter Calibration Data

Current Correct =([0 V Current (mA)]+[5 V Current (mA)]+[10 V Current (mA)]+[15 V Current (mA)]-3

Ammeter	0 V Current (mA)	5 V Current (mA)	10 V Current (mA)	15 V Current (mA)	Current Correct (mA)
1	11.9	16.6	21.4	26.4	11.58
A	-0.6	4.5	9.6	14.7	-0.45
B	-0.2	4.9	10.0	15.1	-0.05

DC Setup

Construct the circuit shown in the Circuit Diagram 2 or in Figure 2. Note that this circuit can be easily gotten from the previous one by removir the white wire between the two current probes from Current Probe A and connecting it to one end of a second 1000 W resistor, so Current probe B is now in series with the new 1000 W resistor. Attach a black wire from the free end of Current Probe A to the black output on the 850 Attach a blue lead from the free end of the new 1000 W resistor to one end of the 3.3 kW resistor and a red lead from the free end of the 3.3 kV resistor to the red output on the 850. Attach a third Voltage Sensor to Analog input C, but leave its free ends unattached for now.



Circuit Diagram 2

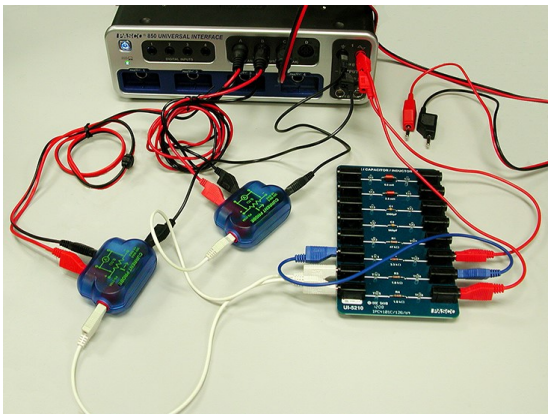


Figure 2: DC Setup

DC Current Procedure

1. Click open the Signal Generator at the left of the screen. Set 850 Output 1 for a DC Waveform and a DC Voltage of 0 V. Click the On button.

2. Click Record (bottom left of screen).

3. Wait several seconds until the measured currents stop varying as the average becomes well defined. Click Stop.

4. The three values should be 0 mA within 0.2 mA or so. The amount they disagree with zero will imply what precision you can expect in your data.

5. Enter the values in row 1 of the table below. Click Delete Last Run.

6. Increase the Signal Generator voltage to 5 V.

7. Enter the three values for current in the first row of the table below.

8. Click Delete Last Run.

9. In the Signal Generator panel, increase the voltage to 10 V and repeat.

Then repeat for 15 V. Turn Signal Generator Off.

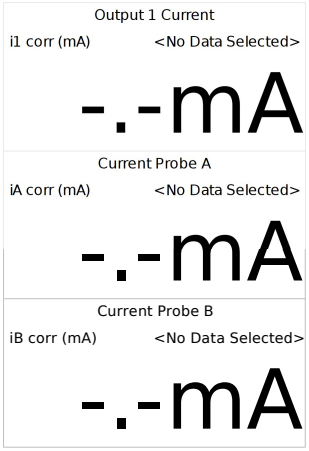
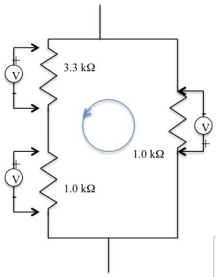


Table II: DC Currents

Output 1 Voltage (V)	i1 (mA)	iA (mA)	iB (mA)	Σi (mA)
0	-0.3	-0.1	0.0	-0.2
5	5.8	1.1	5.0	-0.3
10	12.0	2.3	10.1	-0.4
15	18.3	3.4	15.2	-0.3



Loop Rule Circuit

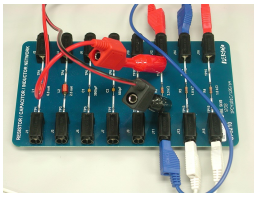


Fig. 3: V across 3.3 kΩ

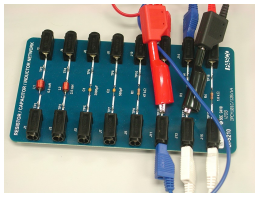


Fig. 4: V across 1st 1 kΩ

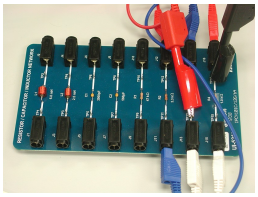


Fig. 5: V across 2nd 1 kΩ

Loop Voltage

Voltage, Ch C (V)<No Data Selected>

-.--V

Table III: Loop Voltages Table

Voltage Out 1	V 3.3k (V)	V 1st 1k (V)	V 2nd 1k (V)	ΣV (V)
5	3.84	1.15	-5.01	-0.02
10	7.69	2.31	-10.02	-0.02
15	11.53	3.46	-15.00	-0.01

DC Voltage Procedure:

To examine the loop rule we must first choose a loop. We take the loop to be as shown in the Loop Rule Circuit above. Note that the circuit is unchanged from that in the DC Current, but now we use the Voltage sensor attached to Analog input C, and move it to the three positions shown in the circuit. We must keep the polarity of the Voltage Sens same with respect to the loop direction to get the signs correctly. I arbitrarily (it doesn't matter) take the arrow showing the loop direction to point from + to – and always atta the red lead from the Voltage Sensor to the + side of the resistor. Of course, if it is really the other way, we will read a negative voltage for that resistor.

1. Attach alligator clips to the banana connectors of the Voltage Sensor attached to Analog input C. Attach the red alligator to TP12 on the Circuit Board as shown in Figure Attach the black alligator to TP 11.
2. Click Record.
3. Click open the Signal Generator. Set the DC Voltage of 850 Output 1 to 5 V. Click On. Read the voltage from the Loop Voltage box and enter it in column 2 (V 3.3k) of tl Loop Voltage Table in the 5 V row. Increase the DC Voltage to 10 V and repeat, entering the value in the 10 V row. Increase the DC Voltage to 15 V and repeat.
4. Move the red alligator to TP 11 (where the black alligator was connected) and move the black alligator to TP 13 on the other side of the 1 kW resistor as shown in Figure
5. Repeat step 3, entering the values in column 3 (V 1st 1k) of the table.
6. Move the red alligator to TP 13 and the black alligator to TP 16 as shown in Figure 5. Note that TP 13 and TP 15 are essentially the same point in the circuit since both a connected to the black Output 1 terminal by a wire (and a 0.1 W resistor).
7. Repeat step 3, entering the values in the 4th column (V 2nd 1k) of the table.
8. Turn the Signal Generator Off.

Conclusions

1. Considering the sum currents in the 5th column of the DC Currents table under the DC Current tab and the uncertainty you estimated in step 4 of the DC Current Procedure, what can you conclude about Kirchhoff's Junction Rule for DC circuits (the currents were too small to permit examination of time varying circuits)?

The sum of the currents is nearly zero within a certain range, validating Kirchhoff's junction rule for DC circuits.

2. Considering the sum voltages in the 5th column of the Loop Voltages table under the DC Voltage tab, what can you conclude about Kirchhoff's Loop Rule for DC circuits?

The total voltage around the loop is approximately zero, within the limits of experimental error, confirming Kirchhoff's Loop Rule for DC circuits.